

Nagarajan Radhakrishnan

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SUMMARY

Ph.D. candidate in Mechanical and Aerospace Engineering (expected to graduate in May 2027) with 7 years of hands-on aerosol instrumentation research, from prototypes at IIT Madras to flight-tested aircraft inlets at Clarkson. Successfully led the CFD design, manufacturing, flight testing and characterization of an aerosol inlet flown on NASA's WB-57 aircraft at Mach 0.7 and 17–18 km. Close the loop between **ANSYS Fluent CFD**, **wind-tunnel and hot-wire experiments**, and **embedded instrumentation**. Collaborated with NOAA, NASA, JPL, NCAR and Harvard.

EDUCATION

Expected May 2027	Ph.D., Mechanical and Aerospace Engineering <i>Clarkson University · Advisor: Prof. Suresh Dhaniyala</i> Thesis: Flow and particle characteristics of a Nested Inlet for Stratospheric Sampling of Aerosol (NISSA)	Potsdam, NY
Jan 2022 – Dec 2023	M.S., Mechanical and Aerospace Engineering <i>Clarkson University</i> Thesis: Study of role of turbulence in anisokinetic aircraft sampling system	Potsdam, NY
2012 – 2016	B.E., Mechanical Engineering <i>Anna University (Sree Sastha Institute of Engineering and Technology)</i>	India

SKILLS

Examples of this work, with figures and interactive demos: nagarajan1295.github.io

CFD (Fluent)	Transonic flow , RANS turbulence modeling, stochastic particle tracking, parametric design and modeling tools.
Instrumentation	Hot-wire anemometry , analog and digital electronics with microcontrollers, DAQ, wind-tunnel testing. Aerosol instruments (APS, SMPS, CPC, optical counters, gravimetric sampling) and low-cost sensors (SPS30, PMS5003).
Programming	MATLAB, Python, NI LabVIEW
Design & build	Autodesk Inventor, SolidWorks, GD&T drafting, CNC machining, FDM 3D printing
Languages	English, Hindi, Tamil, Sanskrit

EXPERIENCE

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Dec 2021 – Present	Graduate Research Assistant <i>Aerosol Transport and Atmospheric Measurement Laboratory, Clarkson University</i> NISSA: Nested Inlet for Sampling Stratospheric Aerosol NASA WB-57, SABRE 2025 campaign - Led CFD design and parametric optimization of a two-stage nested inlet for sampling aerosol (10 nm–10 μm) at Mach 0.7 and about 0.25 atm while keeping inlet turbulence low. SABRE 2025 deployment: in-inlet hot-wire, pressure taps and DPOPS size distributions. Measured turbulence in flight and conducted a ground wind-tunnel test at the same probe station. - Built the flight data-reduction pipeline in MATLAB; calibrated the Teensy 4.1 flight electronics. - Modeled aspiration efficiency of the PurpleAir PA-II-Flex sensor as a function of wind speed.	Potsdam, NY
Aug 2019 – Dec 2021	Project Associate <i>EnERG Lab, Dept. of Applied Mechanics, Indian Institute of Technology (IIT) Madras</i> Led the design, build (ASME GD&T drawings in Inventor) and field test of two optical dust-emission monitoring prototypes at India's largest thermal power plant (4,783 MW).	Chennai, India

2016 – 2019

Earlier roles

Machine Learning Data Associate, Amazon Alexa Data Services (2018–2019); **Service Technician**, OMS Motors (2018); **Business Presentation Specialist**, McKinsey & Company (2016–2017).

PUBLICATIONS AND PRESENTATIONS

Peer-reviewed

2024 Ouimette, J., Arnott, W. P., Laven, P., Whitwell, R., **Radhakrishnan, N.**, Dhaniyala, S., . . . Volckens, J. Fundamentals of low-cost aerosol sensor design and operation. *Aerosol Science and Technology*, 58(1), 1–15. <https://doi.org/10.1080/02786826.2023.2285935>

In progress

Target: Nov 2026 **Radhakrishnan, N.**, Dhaniyala, S. Nested Inlet for Sampling Stratospheric Aerosol (NISSA): Design, development, and evaluation. *To be submitted to Atmospheric Measurement Techniques*.

Target: Dec 2026 **Radhakrishnan, N.**, Dhaniyala, S. Fundamental turbulence measurement in aircraft inlets. *To be submitted to Aerosol Science and Technology*.

In preparation **Radhakrishnan, N.**, Dhaniyala, S. Particle measurements, impact of turbulence in aircraft inlets.

Conference talks and posters

(AAAR Annual Conference unless noted; AAAR = American Association for Aerosol Research)

2025 **Radhakrishnan, N.**, Brock, C., Dhaniyala, S., Keutsch, F., Abou-Rizk, S., Greenberg, M. Performance analysis of Nested Inlet for Sampling Stratospheric Aerosol (NISSA) on a NASA WB-57 high altitude aircraft during SABRE 2025 test flights. *Platform presentation*.

2025 **Radhakrishnan, N.**, Dhaniyala, S. Turbulence measurement in aircraft aerosol inlet. *Clarkson University Research Showcase (RAPS), oral presentation*.

2024 **Radhakrishnan, N.**, Dhaniyala, S. Design and development of Nested Inlet for Sampling Stratospheric Aerosol (NISSA) for effective reduction of turbulence in aircraft aerosol measurements. *Poster*.

2023 **Radhakrishnan, N.**, Dhaniyala, S. Aerosol sampling from aircraft: Impact of turbulence on aspiration and transmission efficiencies. *Platform presentation*.

2022 **Radhakrishnan, N.**, Dhaniyala, S. Aspiration efficiency of PurpleAir PA-II-Flex air quality sensor under varying wind conditions. *Poster*.

2022 **Radhakrishnan, N.**, Dhaniyala, S. Towards minimizing turbulence in aircraft inlets. *Poster*.

HONORS AND AWARDS

2025 **Best Student Oral Platform Presentation** (out of 300 oral presentations), **AAAR**. *Performance analysis of NISSA on a NASA WB-57 high altitude aircraft during SABRE 2025 test flights*

2025 **Best Graduate Oral Presentation, Clarkson RAPS**. *Turbulence measurement in aircraft aerosol inlet*

2025 **3rd Place, Project Pitch to a General Audience, Clarkson RAPS**

2022 **Best Student Poster, AAAR**. *Towards minimizing turbulence in aircraft inlets*

Press Featured by the NOAA Climate Program Office and Clarkson University news.

TEACHING AND LEADERSHIP

Teaching assistant **ME 301 Experimental Methods for Aerospace and Mechanical Engineering** (junior-level laboratory). Run the wing pressure lab and other lab experiments, teaching students to set up, measure and analyze.

Teaching assistant **ME 310 Thermodynamic System Engineering** (junior/senior-level course). Course support, exam preparation, student consultation.

2022 – Present **President, AAAR Student Chapter**. Run workshops and the yearly AS&T lecture series.

2022 – Present **Advisor and Media Chair, Clarkson Garden Club**. Run the university greenhouse, garden and advice undergraduate students.